Collagen Synthesis

Learning Objectives

By the end of the lecture the student shall be able to

- Enumerate the steps of collagen synthesis
- Explain the sites and importance of collagen
- Enumerate the steps of collagen degradation

INTRODUCTION

Collagen is a family of highly developed fibrous proteins found in all multicellular animals. It is the most abundant proteins found in mammals, constituting 25 percent of the total protein mass.

Collagen is the main fibrous component of skin, bone ,tendon, cartilage and periodontium.

It comprises about 90% of the organic matrix of the bone. ***Collagen is a natural protein that provides our bodies with structural support***.

It seems collagen evolved early in evolution and its composition and structure have changed relatively little suggesting its design has reached optimum efficiency.

EXTRA CELLULAR TISSUE MATRIX

(PERIODONTIUM)

FIBROUS + GROUND STR.

Collagen
Elastin
Oxytalin
Elauin
(PROVIDES TENSILE STRENGTH)

Non Collagenous proteins
Proteoglycans, Hyaluronan
Lipids , Water
(Provides Compressive Strength)

Collagen

- It is the most abundant protein in the body.
- It is long, rigid structure in which three polypeptides are wound around one another in a rope like fashion.
- These polypeptides are called α-helix.
- They are arranged in a triple helix.
- They are found everywhere in the body, but their type is dictated by their structural role in a particular organ.

- Example:

1. Gel- extracellular matrix or vitreous humor of eye.
2. Tight bundles- Tendons
3. Stacked- as in Cornea
4. Fibers arranged at an angle- Bones

Types of Collagen

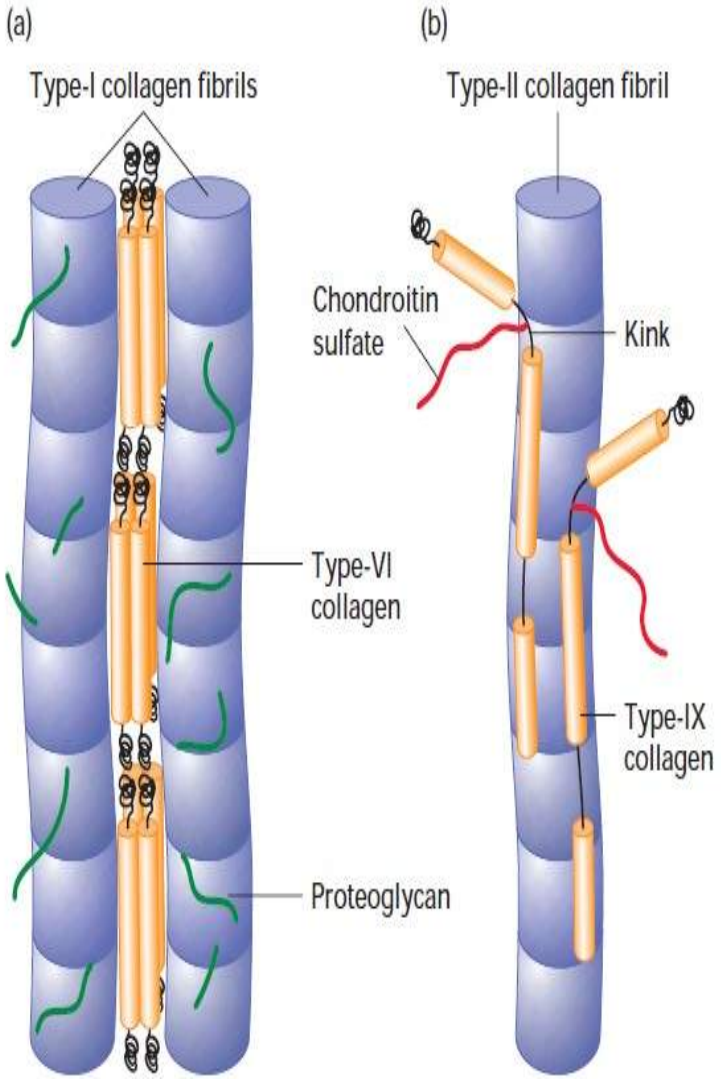
Collagens represent a large family of proteins and 19 different collagen types have been described so far.

Many genetically, chemically, and immunologically distinct types of collagen have been identified

VARIATIONS ARE DUE TO .....

1. Differences in the assembly of basic polypeptide chains
2. Different lengths of the helix
3. Various interruptions in the helix and
4. Differences in the terminations of the helical domains

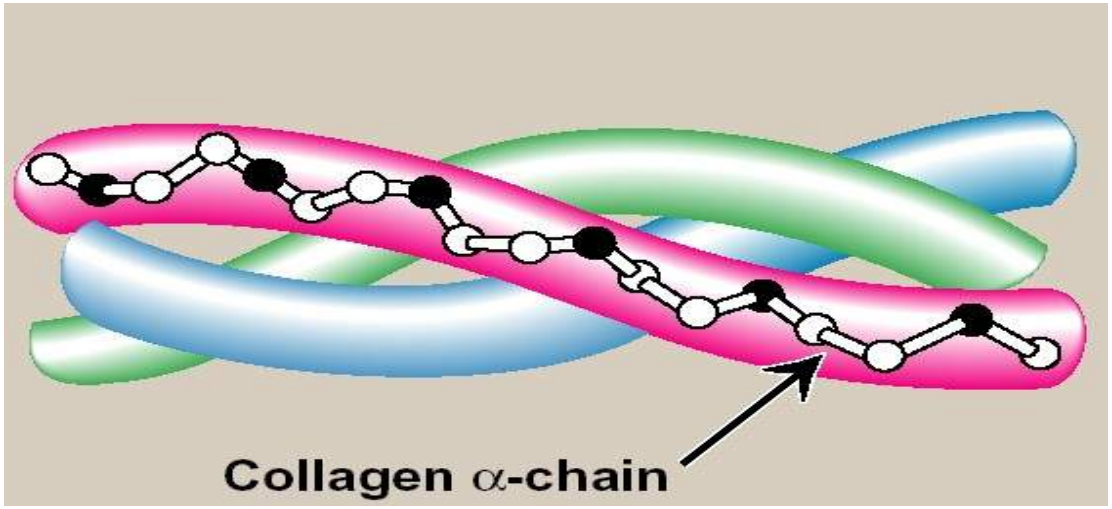
Types of Collagen



Type	Genes	Tissue
I	*COL1A1, COL1A2*	Most connective tissues, including bone
II	*COL2A1*	Cartilage, vitreous humor
III	*COL3A1*	Extensible connective tissues such as skin, lung, and the vascular system
IV	*COL4A1–COL4A6*	Basement membranes
V	*COL5A1–COL5A3*	Minor component in tissues containing collagen I
VI	*COL6A1–COL6A3*	Most connective tissues
VII	*COL7A1*	Anchoring fibrils
VIII	*COL8A1–COL8A2*	Endothelium, other tissues
IX	*COL9A1–COL9A3*	Tissues containing collagen II
X	*COL10A1*	Hypertrophic cartilage
XI	*COL11A1, COL11A2, COL2A1*	Tissues containing collagen II
XII	*COL12A1*	Tissues containing collagen I
XIII	*COL13A1*	Many tissues
XIV	*COL14A1*	Tissues containing collagen I
XV	*COL15A1*	Many tissues
XVI	*COL16A1*	Many tissues
XVII	*COL17A1*	Skin hemidesmosomes
XVIII	*COL18A1*	Many tissues (eg, liver, kidney)
XIX	*COL19A1*	Rhabdomyosarcoma cells

STRUCTURE OF COLLAGEN

- **Amino Acid Sequence**
- **Triple- helical structure**
- **Hydroxyproline & Hydroxylysine**
- **Glycosylation**



Amino Acid Sequence

Collagen is a glycoprotein containing galactose and glucose as the carbohydrate content.

Repeating unit is Gly-X-Y

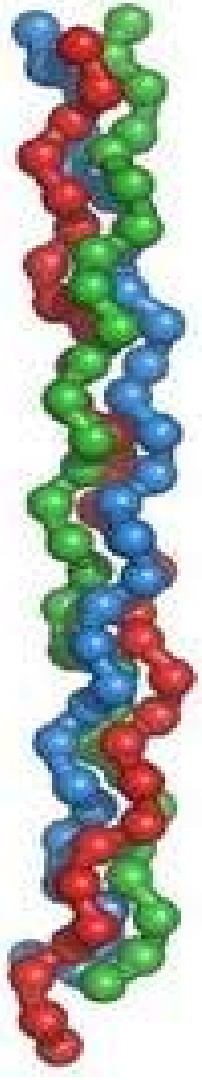
Proline frequently occupies the X position.

Hydroxyproline and hydroxylysine are two unique amino acids. Vertebrate collagens exhibit the two of them in the Y positions

Each collagen molecule is stabilized through lysine derived intra and intermolecular cross links.

Each α chain comprises of around 1056 amino acids.

There are around 3 amino acids per turn.



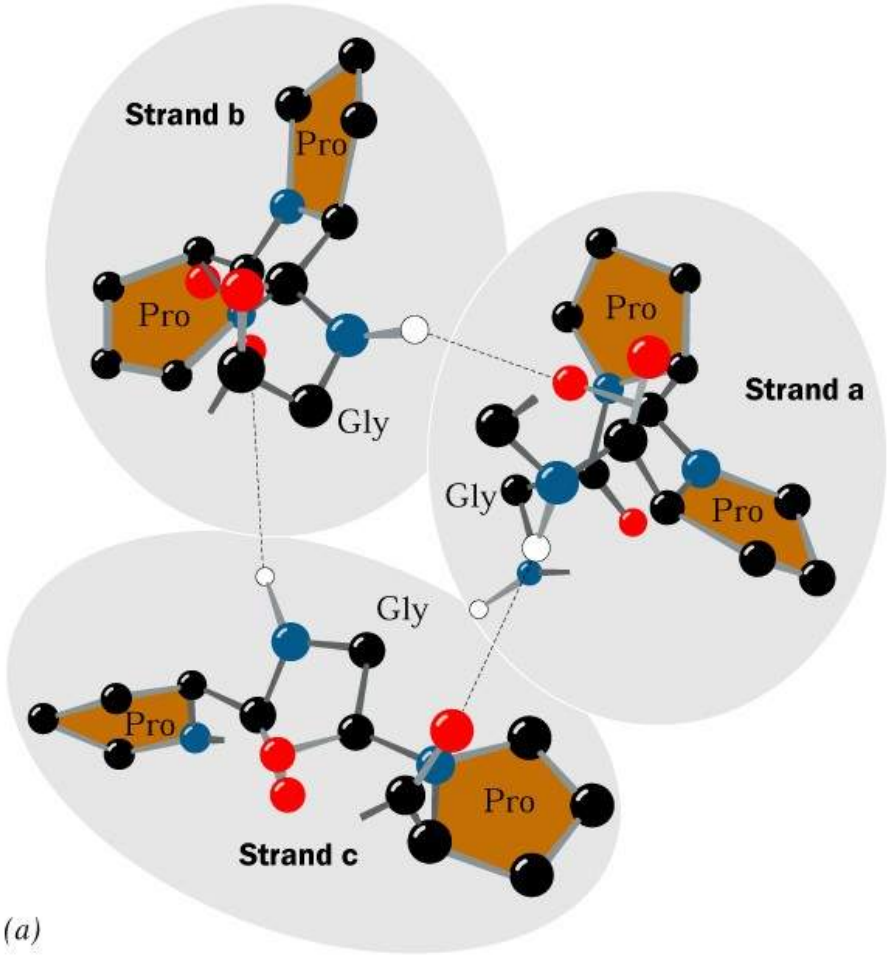
Sequence of Amino Acids in Collagen



The Collagen Triple Helix (Tropocollagen)

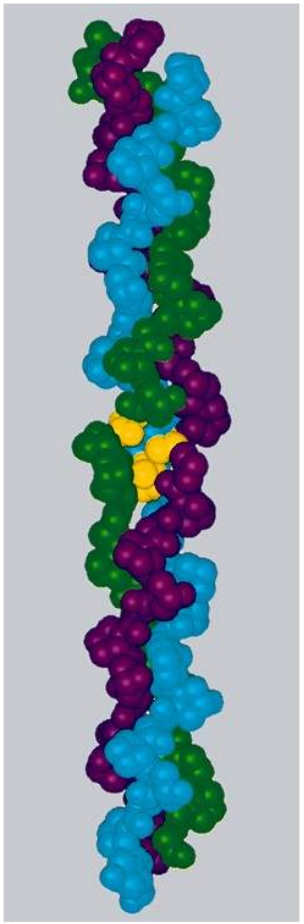


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Interactions between strands

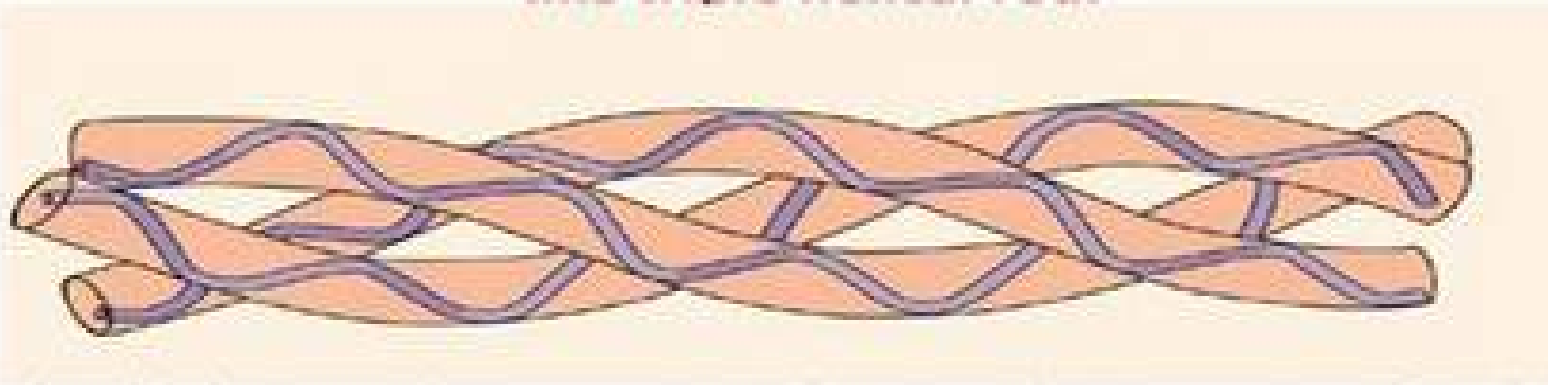


Courtesy of Helen Berman, Rutgers University.

Tropocollagen with Gly → Ala substitution (yellow)

G.N. Ramachandran, 1955

The α chains are left handed helices that wrap around each other into a right handed rope like triple helical rod.



Each such helix is around 1.4 nanometers in diameter and 300 nanometers in length

THE TRIPLE HELIX MAY BE OF A CONTINUOUS STRETCH OR IT MAY BE INTERRUPTED BY NON COLLAGENOUS ELEMENTS.

Glycine occupies every third position in the repeating amino acid sequence

GLYCINE-X-Y

- Glycine is one - third of total amino acid content of collagen followed by hydroxyproline and proline account for another one-third of amino acid content of collagen.

- Proline - facilitate the formation of helical conformation of α- chain, because its ring structure causes kink in the peptide chain.

- Glycine- found in every third position of the polypeptide chain. It fits into the restricted spaces where the three chains of the helix come together.

- Glycine is the part of the repeating sequence.

  Gly- X-Y

- X- is frequently proline
- Y- hydroxy proline or hydroxylysine.

Triple- helical structure:

- Amino acids side chains are on the surface of the triple helical molecule.

- This allows bond formation between the exposed R- groups of neighboring collagen monomers- This leads to aggregation into fibrils.

Hydroxyproline & Hydroxylysine:

- Hydroxylation of Proline & lysine residues after their incorporation into the polypeptide chains.

- Thus called post translational modification.

- Causes stabilization of triple helical structure.

Glycosylation:

- Hydroxyl group of hydroxylysine residues of collagen are enzymatically glycosylated.

- Most commonly glucose and galactose are attached.

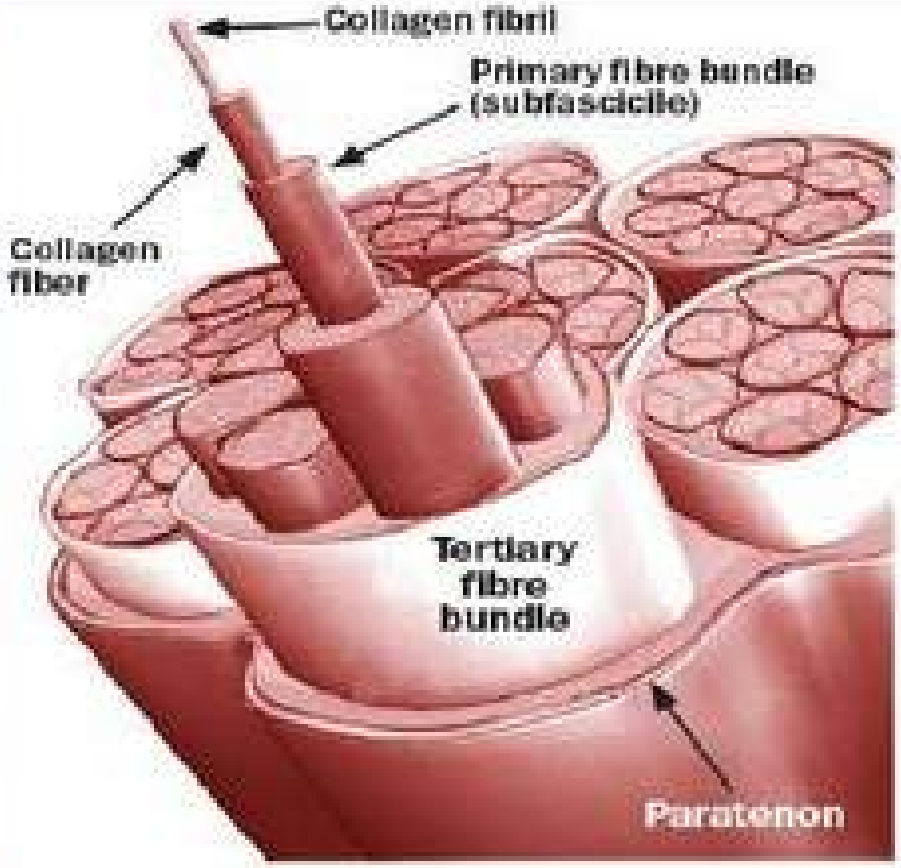
COLLAGEN IN PERIODONTIUM

The collagen of PDL is largely Type I , with lesser amounts of type III , IV , VI and XII .

Fibril Collagens include type I , III , V that generally co distributes with collagen type I. Higher portion of collagen III usually seen in foetal tissues.

Collagen Type IV does not form fibrils and is found in basement membranes of neurovascular bundles and other PDL cells. Though this collagen is specifically associated with basement membranes , it also plays role in maintaining the elastic system of the vasculature of the gums.

Collagen fibers of the periodontium (particularly Type I) provide the structural requirements to withstand intrusive forces of mastication (tooth support) and also to accommodate growing tooth in mammals.



BIOSYNTHESIS OF COLLAGEN

Sites For The Synthesis of Collagen :



1.Mesenchymal Cells & Their Derivatives



FIBROBLASTS (major cells)



Chondrocytes



Osteoblasts



Odontoblasts



CEMENTOBLASTS

2.Other Cells : Epithelial cells. Endothelial cells. Muscle cells. Schwann cells.

BIOSYNTHESIS OF COLLAGEN

Precursors:

- Collagen is one of the proteins that functions outside the cell.

- Polypeptide Precursors of the collagen molecule are formed in Fibroblast, osteoblasts and chondroblasts.

- These are secreted into the extracellular matrix.

1. **Formation of Pro- α-chains:**

- Pre-pro α-chains- contain a special amino acid sequence at their N-terminal.
- This sequence acts as a signal that the newly synthesized polypepetide is destined for function out side the cell.

- This sequence facilitate the binding of ribosomes to the rough endoplasmic reticulum (RER), and direct the Pre-pro α-chain into the lumen of the RER.

- This sequence is cleaved in the lumen of RER and after its cleavage Precursor of collagen is formed.

- This precursor is called Pro α-chain.

2. **Hydroxylation:**

- Processing of Pro α-chains occur by a number of enzymic steps in the lumen of **RER**, while the polypeptides are still being synthesized.

- Proline and lysine residues are hydroxylated.

- This reaction requires O2 and vitamin C.

- Enzymes are prolyl hydroxylase and lysyl hydroxylase.

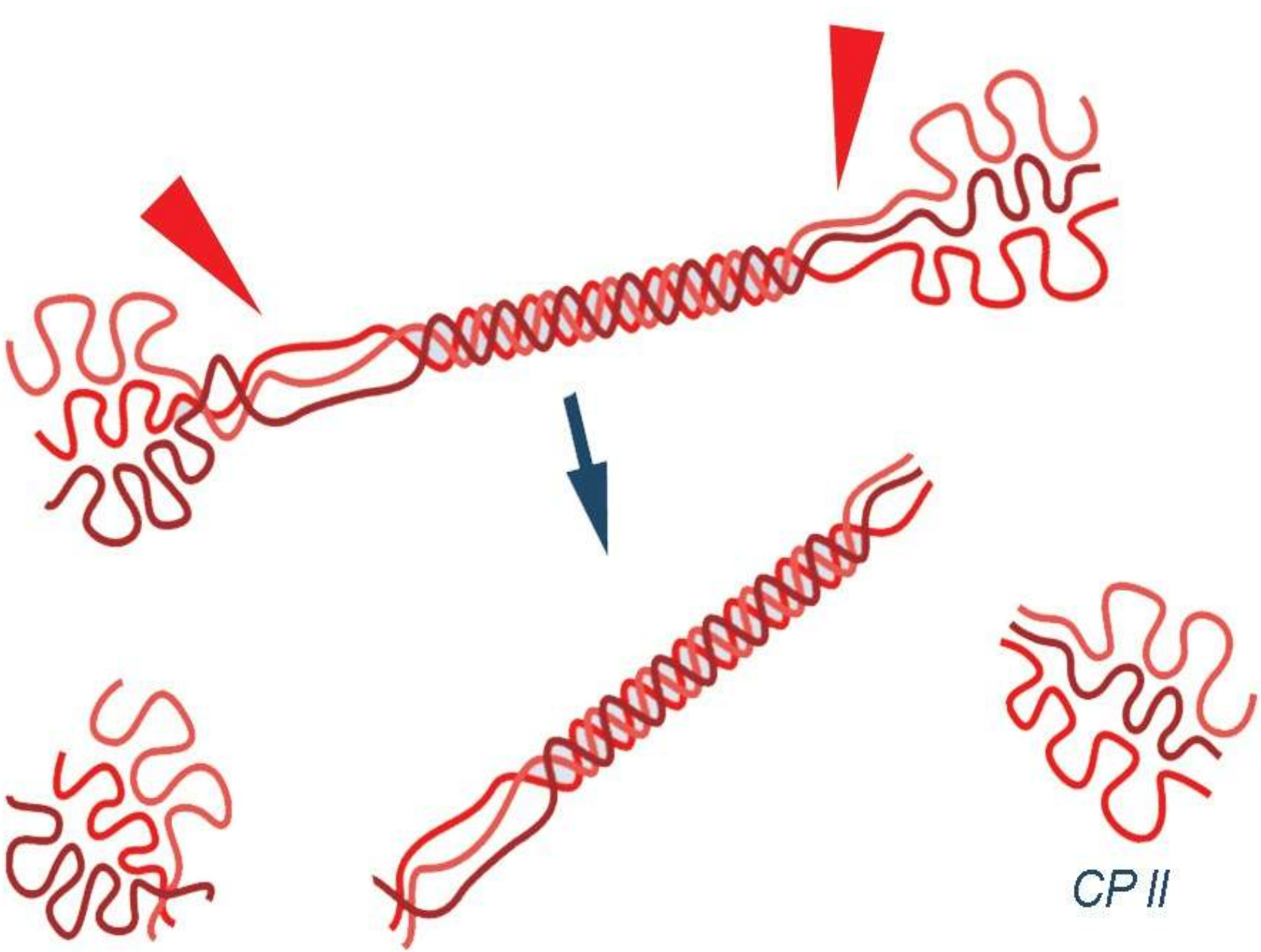
- In Vit C deficiency, collagen fibers cannot cross link- and tensile strength is decreased (scurvy).

3. **Glycosylation:**

- Modified by glycosylation with glucose or galactose residues.

4. **Assembly and Secretion:**

- After hydroxylation and glycosylation- Pro α-chains are converted to Pro-collagen.
- Pro-collagen has a central region of triple helix and its ends have non-helical regions of amino and carboxyl terminal extensions .
- These extensions are called Propeptides.


CP II

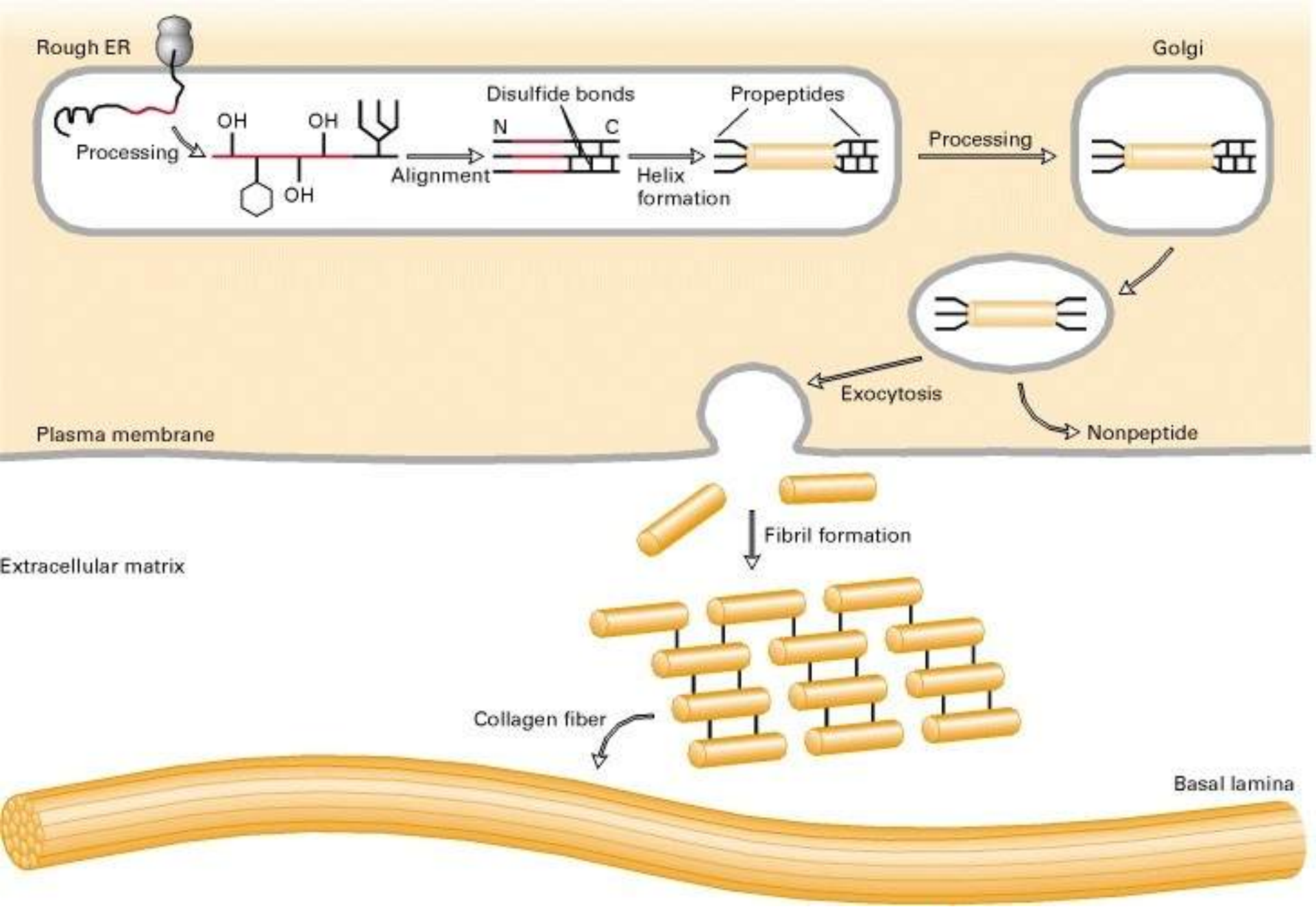
- In the formation of procollagen interchain disulfide bonds are formed between the C-terminal extensions of the pro α-chains.

- This alignment of pro α-chains is favorable for helix formation.

- Then pro-collagen chains are translocated to Golgi- apparatus.

- In the golgi they are packaged in secretory vesicles.

- These vesicles fuse with the membrane and release the **pro-collagen molecule into the extracellular space**.

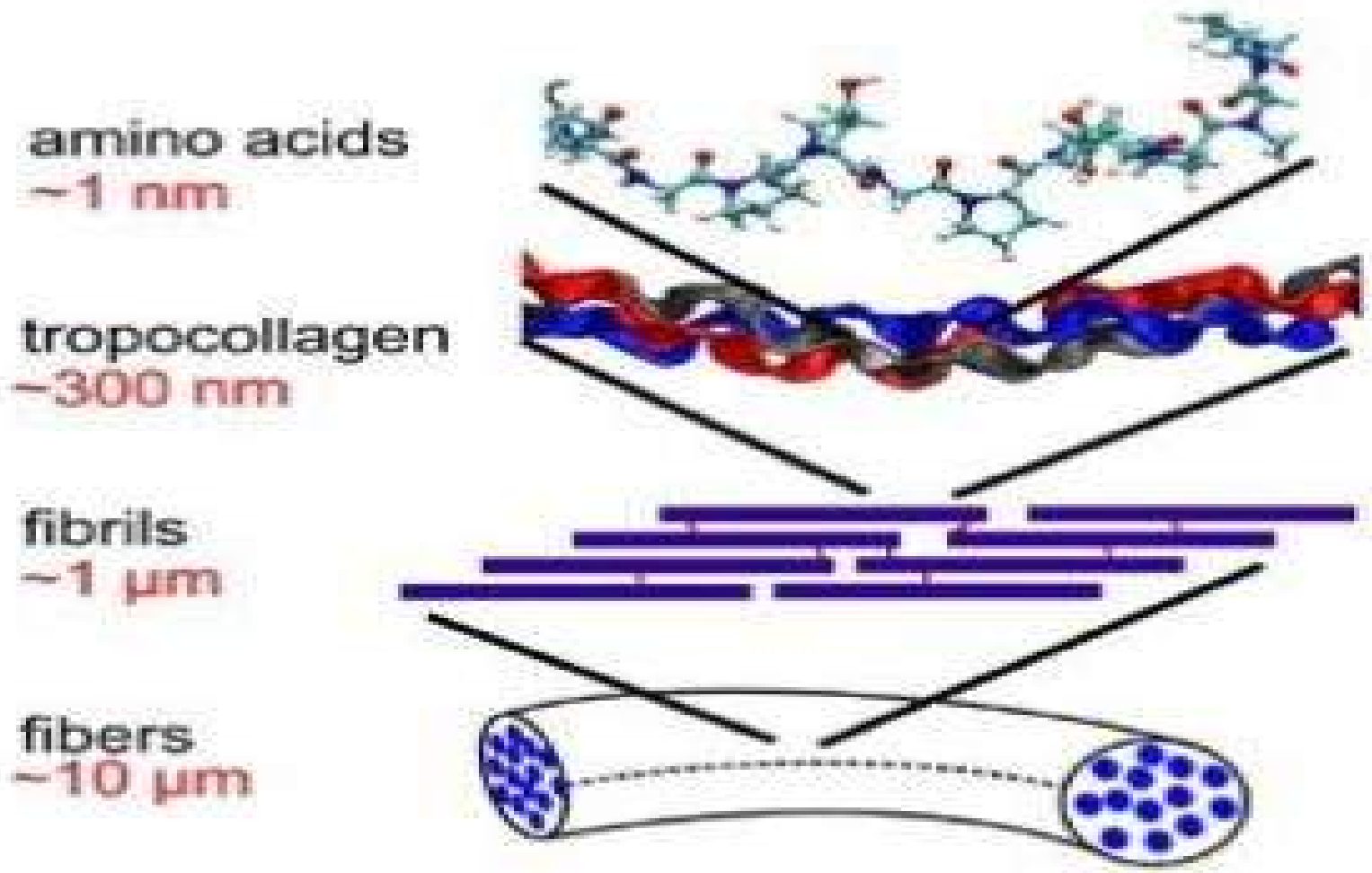

Rough ER
Processing
OH
OH
OH
Alignment
Disulfide bonds
N
C
Helix formation
Propeptides
Processing
Golgi
Exocytosis
Nonpeptide
Plasma membrane
Extracellular matrix
Fibril formation
Collagen fiber
Basal lamina

5. **Extracellular cleavage of Procollagen molecules**:

- After their release the Procollagen molecules are cleaved by N- and C Procollagen peptidases.
- These remove the terminal Propeptides.
- Triple helical structure is released as Tropocollagen.

6. **Formation of collagen fibrils:**

- Tropocollagen spontaneously associate with each other and form collagen fibrils.


amino acids
~1 nm
tropocollagen
~300 nm
fibrils
~1 µm
fibers
~10 µm

7. **Cross-link formation:**

- The fibrils that are formed become a substrate for lysyl oxidase.
- It contains copper.
- It oxidatively deaminates lysyl and hydroxlysyl residues in collagen.
- Reactive aldehydes- Allysine and hydroxylysine are formed.
- These aldehydes the react with the neighboring lysyl and hydroxlysyl and covalent cross links are formed.
- This cross-linking leads to the formation of mature collagen.

Steps involved in collagen biosynthesis

Rough Endoplasmic Reticulum

- Synthesis of preprocollagen
- Insertion of procollagen molecule into the lumen of ER.

Lumen of ER:

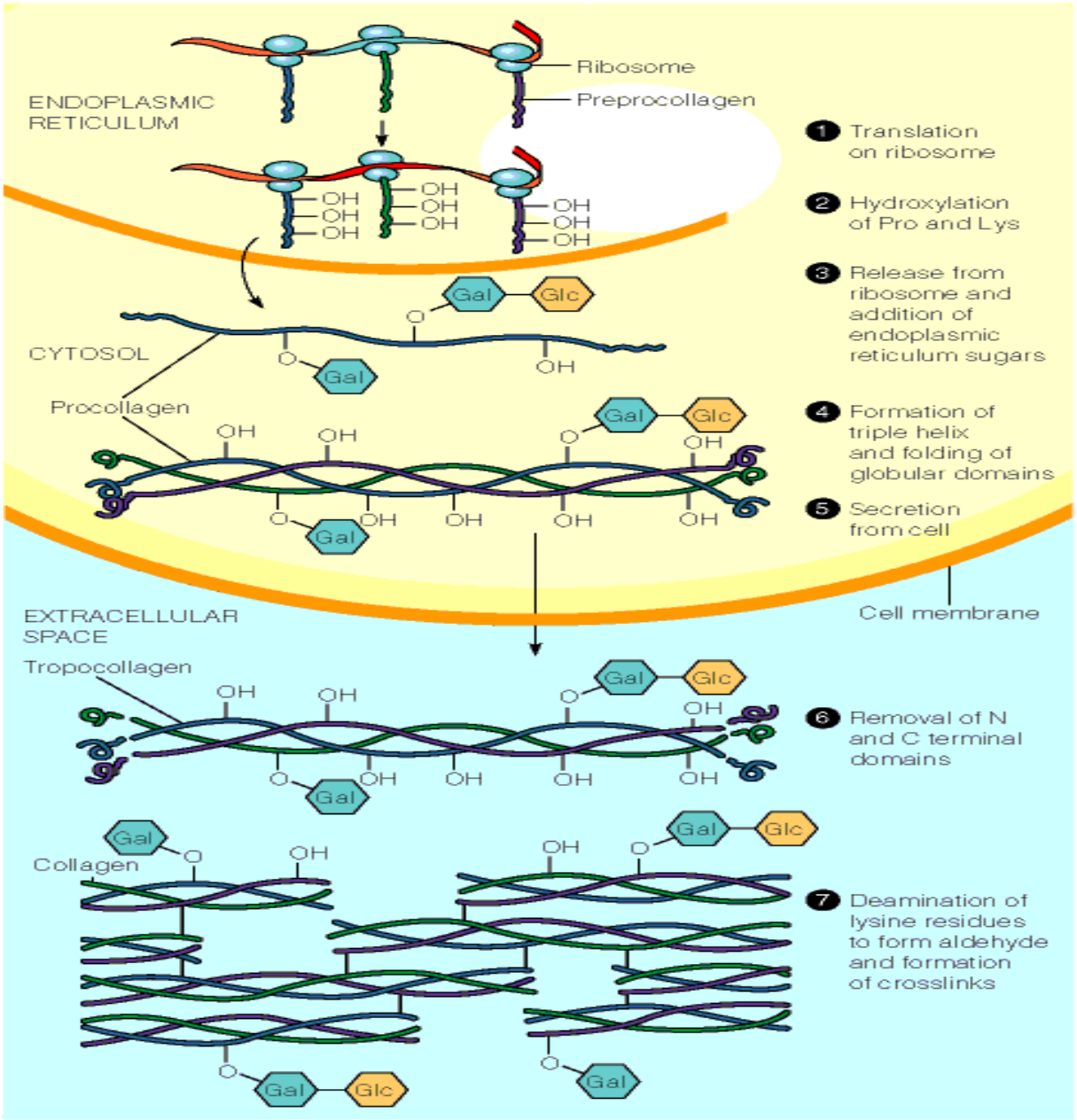
- Hydroxylation of proline and lysine residues.
- Glycosylation of selected hydroxylysine residues.

Lumen of ER and Golgi apparatus:

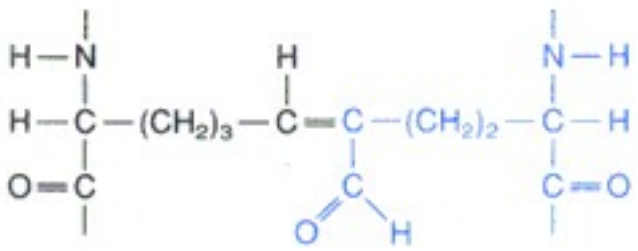
- Self assembly of tropocollagen molecule (disulfide bond formation).

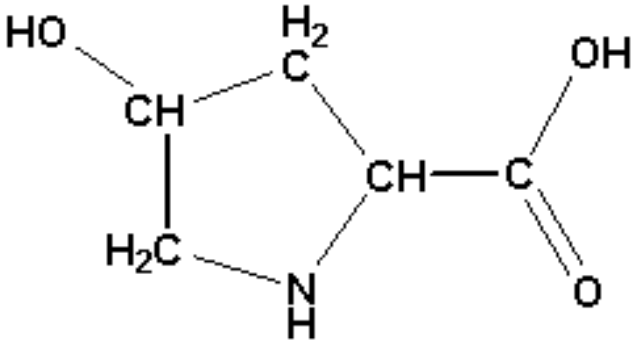
Secretory vesicles:

Biosynthesis and Assembly of Collagen



Post-Translational Modifications in Collagen





Hydroxyproline

Collagen contains unusual, oxidized and crosslinked lysine residues.

Lysyl oxidase is the enzyme that oxidizes lysine residues to the aldehyde allysine, which then forms the crosslinks.

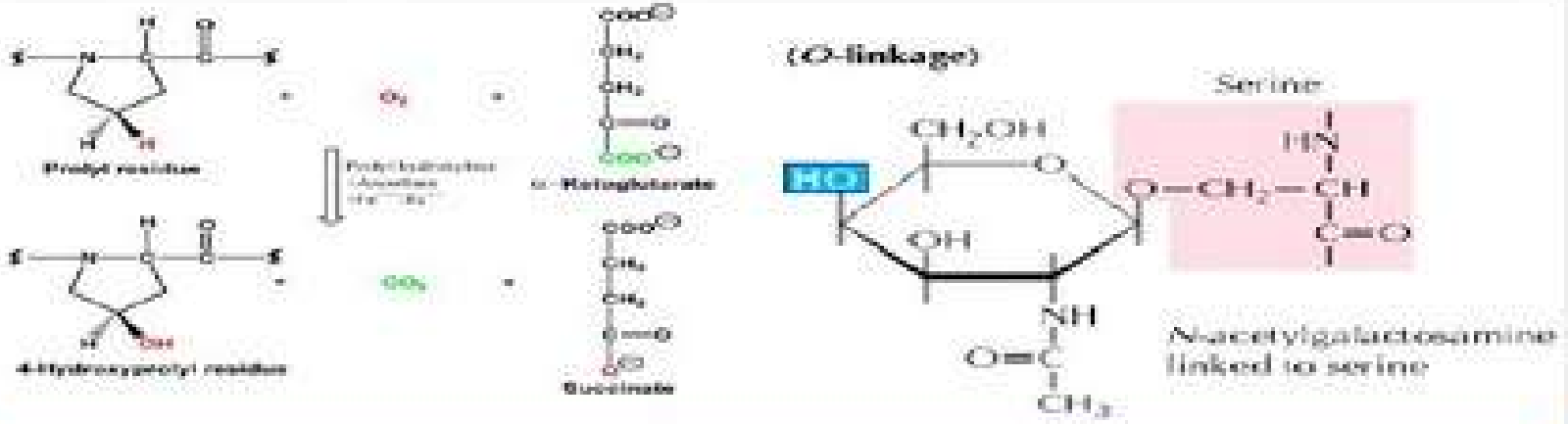
Hydroxyproline is also found in collagen. (Some lysine residues also hydroxylated.) The enzyme required for hydroxylation of proline residues is prolyl hydroxylase, a vitamin C-dependent enzyme.

Scurvy is caused by reduced hydroxyproline in collagen as a result of vitamin C deficiency.

Significance of Hydroxylation & Glycosylation

Triple Helical Type 1 collagen, with complete hydroxylation of Prolyl has a **Tm of 39*C.** However underhydroxylated molecules melt at lower temperatures i.e the Tm is around 25*C.

Therefore Prolyl hydroxylation is essential for **collagen thermal stability** at physiologic temperature. While underhydroxylated molecules are denatured and degraded.



Hydroxylysine residues on the other hand , serve as specific glycosylation sites and form more stable cross links than lysine.

Regulation of Collagen Biosynthesis

Collagen biosynthesis is tightly regulated during normal development & homeostasis in a cell & tissue specific manner .

GENE TRANSCRIPTION

The changes at the gene transcription stage are reflected by dec in the mRNA levels and unstable mRNA.

Collagen genes contain CIS-regulatory sequences , promoters and enhancers.

Gene transcription is regulated by binding of protein transcription factors to the sequences.Eg. TGF-β activates collagenase gene transcription through regulatory DNA sequence that binds to NF-1.

POST TRANSLATIONAL MODIFICATION

Post Translationally Col synthesis is regulated by the extent of prolyl hydroxylation .

Under Hydroxylation results in dec stability of the collagen molecule which is then degraded .

Correct amount of hydroxylation results in the production of stable and strong collagen fibers.